

## Quiz 2 Section 1 Detailed Solutions and Grading

A continuous random variable  $Y$  has the probability density function

$$f(y) = \begin{cases} k(1-y), & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Each question is worth **5 points** (total 20 points).

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### 1. Find the constant $k$ .

We require

$$\int_{-\infty}^{\infty} f(y) dy = 1.$$

Since  $f(y) = k(1-y)$  on  $[0, 1]$ :

$$1 = \int_0^1 k(1-y) dy = k \left[ y - \frac{y^2}{2} \right]_0^1 = k \left( 1 - \frac{1}{2} \right) = \frac{k}{2}.$$

Hence,

$k = 2.$

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### 2. Determine the cumulative distribution function $F(y)$ .

$$F(y) = P(Y \leq y) = \int_0^y 2(1-t) dt = 2 \left[ y - \frac{y^2}{2} \right]_0^y = 2y - y^2.$$

Thus,

$$F(y) = \begin{cases} 0, & y < 0, \\ 2y - y^2, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

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**3. Compute  $P(0.25 \leq Y \leq 0.75)$ .**

$$P(0.25 \leq Y \leq 0.75) = F(0.75) - F(0.25).$$

Compute:

$$F(0.75) = 2(0.75) - (0.75)^2 = 1.5 - 0.5625 = 0.9375,$$

$$F(0.25) = 2(0.25) - (0.25)^2 = 0.5 - 0.0625 = 0.4375.$$

$$P(0.25 \leq Y \leq 0.75) = 0.9375 - 0.4375 = 0.5.$$

$P(0.25 \leq Y \leq 0.75) = 0.5.$

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**4. Find the mean and variance of  $Y$ .**

**Mean:**

$$E[Y] = \int_0^1 y \cdot 2(1-y) dy = 2 \int_0^1 (y - y^2) dy = 2 \left[ \frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 = 2 \left( \frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}.$$

$E[Y] = \frac{1}{3}.$

**Second moment:**

$$E[Y^2] = \int_0^1 y^2 \cdot 2(1-y) dy = 2 \int_0^1 (y^2 - y^3) dy = 2 \left[ \frac{y^3}{3} - \frac{y^4}{4} \right]_0^1 = 2 \left( \frac{1}{3} - \frac{1}{4} \right) = \frac{1}{6}.$$

**Variance:**

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = \frac{1}{6} - \left( \frac{1}{3} \right)^2 = \frac{1}{18}.$$

$\text{Var}(Y) = \frac{1}{18}.$

Numerical forms:

$$E[Y] \approx 0.3333, \quad \text{Var}(Y) \approx 0.0556.$$

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| Question | Key Steps | Points Allocation |
|----------|-----------|-------------------|
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## Grading Guideline (5 points each)

| Question                         | Key Steps                                                                            | Points Allocation |
|----------------------------------|--------------------------------------------------------------------------------------|-------------------|
| <b>1. Find <math>k</math></b>    | Setup normalization (1),<br>evaluate integral (2), solve<br>$k = 2$ (2)              | <b>5 pts</b>      |
| <b>2. Find <math>F(y)</math></b> | Piecewise setup (1), integrate<br>correctly (3), present final (1)                   | <b>5 pts</b>      |
| <b>3. Find probability</b>       | Setup integral or CDF<br>difference (1), correct<br>arithmetic (3), final result (1) | <b>5 pts</b>      |
| <b>4. Mean &amp; Variance</b>    | Setup $E[Y]$ (1), evaluate (2),<br>compute $E[Y^2]$ (1), correct<br>variance (1)     | <b>5 pts</b>      |

**Partial credit rules:** - Arithmetic slips but correct setup: deduct 1–2 pts. - Missing piecewise cases in  $F(y)$ : deduct up to 2 pts. - Conceptual misunderstanding (wrong limits, ignoring normalization): major deduction.

## Final Answers Summary

$$\begin{aligned}
 k &= 2, \\
 F(y) &= \begin{cases} 0, & y < 0, \\ 2y - y^2, & 0 \leq y \leq 1, \\ 1, & y > 1, \end{cases} \\
 P(0.25 \leq Y \leq 0.75) &= 0.5, \\
 E[Y] &= \frac{1}{3}, \\
 \text{Var}(Y) &= \frac{1}{18}.
 \end{aligned}$$